Phishing Email Detection & Awareness System

Task 2 – Email Threat Analysis & Risk Assessment

Target Area:

Organizational Email Communication Security

Assessment Scope:

Phishing Detection, Risk Classification & Awareness Framework

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Table of Contents

Table of Contents..... 2
Executive Summary..... 3
Disclaimer..... 4
Introduction..... 5
Objective..... 5
What is Phishing?..... 5
Types of Phishing Attacks.....6-7
Tools Used.....7
Email Analysis Methodology.....7-14
Sample Email Analysis.....15
Email 1 – Netflix Phishing.....15-16
Email 2 – American Express Phishing.....16-17
Email 3 – Business Email Compromise (Maersk).....17-18
Email 4 – Advance Fee Scam.....18-19
Email 5 – Malware Delivery (Voicemail Scam).....19-20
Phishing Indicators Identified.....21-22
Impact on Organization.....23
Prevention Guidelines.....24
Awareness Guidelines for Employees.....24
Conclusion.....25
References.....26

Executive Summary

This report presents the analysis and assessment of multiple email samples to identify phishing threats and evaluate associated security risks. Phishing remains one of the most prevalent cyber threats targeting organizations worldwide, exploiting human behavior through deceptive communication techniques.

The objective of this assessment was to examine suspicious email samples, identify phishing indicators such as spoofed domains, malicious links, suspicious attachments, urgency tactics, and grammatical inconsistencies, and classify each case based on risk severity.

A structured methodology was followed, including sender domain verification, link inspection, attachment analysis, header review, and psychological manipulation assessment. External threat intelligence tools such as Google Safe Browsing and VirusTotal were utilized to validate suspicious URLs and evaluate potential malicious activity.

The findings indicate that several analyzed emails were high-risk phishing attempts, including credential harvesting attacks, financial impersonation fraud, business email compromise (BEC), advance-fee scams, and malware delivery attempts.

The report further outlines the potential organizational impact of phishing attacks, including financial losses, data breaches, reputational damage, and regulatory consequences. Finally, technical prevention measures and employee awareness guidelines are provided to strengthen organizational resilience against phishing threats.

This assessment highlights the critical importance of combining technical controls with continuous user awareness to effectively mitigate phishing risks within modern digital environments.

Disclaimer

This report has been prepared solely for educational and internship purposes as part of the Cyber Security Internship program at Future Interns.

All email samples analyzed in this document are used strictly for cybersecurity awareness and academic evaluation. No real systems were targeted, accessed, or harmed during the preparation of this report.

The phishing examples discussed are publicly available samples or simulated scenarios intended to demonstrate common attack techniques. Any resemblance to actual organizations or systems is purely for analytical and educational purposes.

The author assumes no responsibility for misuse of the information contained in this report. This document is intended to promote cybersecurity awareness, risk understanding, and preventive best practices against phishing attacks.

Introduction

In recent years, cyber attacks have increased significantly across industries, targeting organizations of all sizes. Among various cyber threats, email phishing has emerged as one of the most common and dangerous attack methods. Attackers use deceptive emails to trick employees into revealing sensitive information such as login credentials, financial details, and confidential business data.

Organizations face serious challenges due to phishing attacks, including financial losses, data breaches, reputational damage, and operational disruption. Since email remains a primary communication channel in businesses, even a single successful phishing attempt can compromise critical systems and sensitive information. Therefore, identifying phishing emails and strengthening user awareness has become an essential component of organizational cybersecurity strategy.

Objective

The primary objective of this task is to analyze email samples to identify potential phishing indicators and classify them based on risk level (Safe, Suspicious, or Phishing).

This report aims to:

- Detect common phishing characteristics such as spoofed sender addresses, fake domains, malicious links, and urgent language.
- Evaluate the potential risk associated with each email.
- Explain the identified threats in clear, business-friendly language.
- Provide practical prevention and awareness guidelines to reduce phishing risks within organizations.

What is Phishing?

Phishing is a cyber attack in which criminals send fake emails to trick people into sharing sensitive information. These emails are designed to look real and often appear to come from trusted organizations such as banks, payment services, or company departments.

In many phishing attacks, the sender address is spoofed. This means the attacker makes the email look like it is coming from a legitimate source, even though it is fake. For example, the email may look like it is from “support@paypal.com,” but the actual domain is slightly changed to deceive the recipient.

Phishing emails often contain links that redirect users to fake login pages. These fake websites are carefully designed to look identical to official websites. When users enter their usernames and passwords on these pages, the information is captured by the attacker.

The main objective of phishing is credential stealing. Once attackers obtain login credentials, they can access accounts, steal financial data, misuse company systems, or perform further cyberattacks. Because phishing relies on human trust and urgency, it remains one of the most successful and dangerous cyber threats faced by organizations today.

Types of Phishing Attacks

Phishing attacks can take different forms depending on the target and communication method. The most common types are:

1 Email Phishing

This is the most common type of phishing attack. Attackers send bulk fake emails to many users, pretending to be from trusted organizations such as banks or online services. These emails usually contain malicious links or attachments designed to steal login credentials or financial information.

2 Spear Phishing

Spear phishing is a targeted attack aimed at a specific individual or organization. Unlike general email phishing, the attacker personalizes the message using the victim's name, job role, or company details to make it more convincing and harder to detect.

3 Whaling

Whaling is a specialized form of spear phishing that targets high-level executives such as CEOs, CFOs, or senior managers. These attacks often attempt to steal large financial transactions or confidential corporate data.

4 Smishing

Smishing refers to phishing attacks conducted through SMS (text messages). Victims receive fake messages containing malicious links or urgent requests, such as prize claims or account verification alerts.

5 Vishing

Vishing (voice phishing) involves attackers making fraudulent phone calls while pretending to be bank officials, IT support, or government representatives. The goal is to convince the victim to share sensitive information over the phone.

Tools Used

The following tools were used to Analyze and classify suspicious URLs for phishing detection.

1 Google Safe Browsing (Transparency Report)

Google Safe Browsing was used to check whether a website is currently flagged as unsafe. It helped identify whether the site contains harmful content or attempts to trick users into sharing personal information.

2 Virus Total

Virus Total was used to scan suspicious URLs against multiple cybersecurity engines. The detection ratio (e.g., 16/94 flagged) helped classify links into High, Medium, and Low risk categories based on vendor analysis.

3 Google Docs / MS Word

Google Docs / MS Word was used to structure the report professionally, organize findings, insert screenshots, and prepare the final PDF submission.

Email Analysis Methodology

1 Check Sender Email Address

The sender email shown is:

info@confirm.com

Analysis:

- Official Netflix emails are sent from domains like:
  - @netflix.com
- The sender domain **confirm.com** is unrelated to Netflix.

- This is a classic example of **brand impersonation using domain mismatch**.

Red Flag Identified: Sender domain does not match official company domain.

2 Verify Domain Legitimacy

The email contains a visible link:

https://login-memberarea.netflix.com

Analysis:

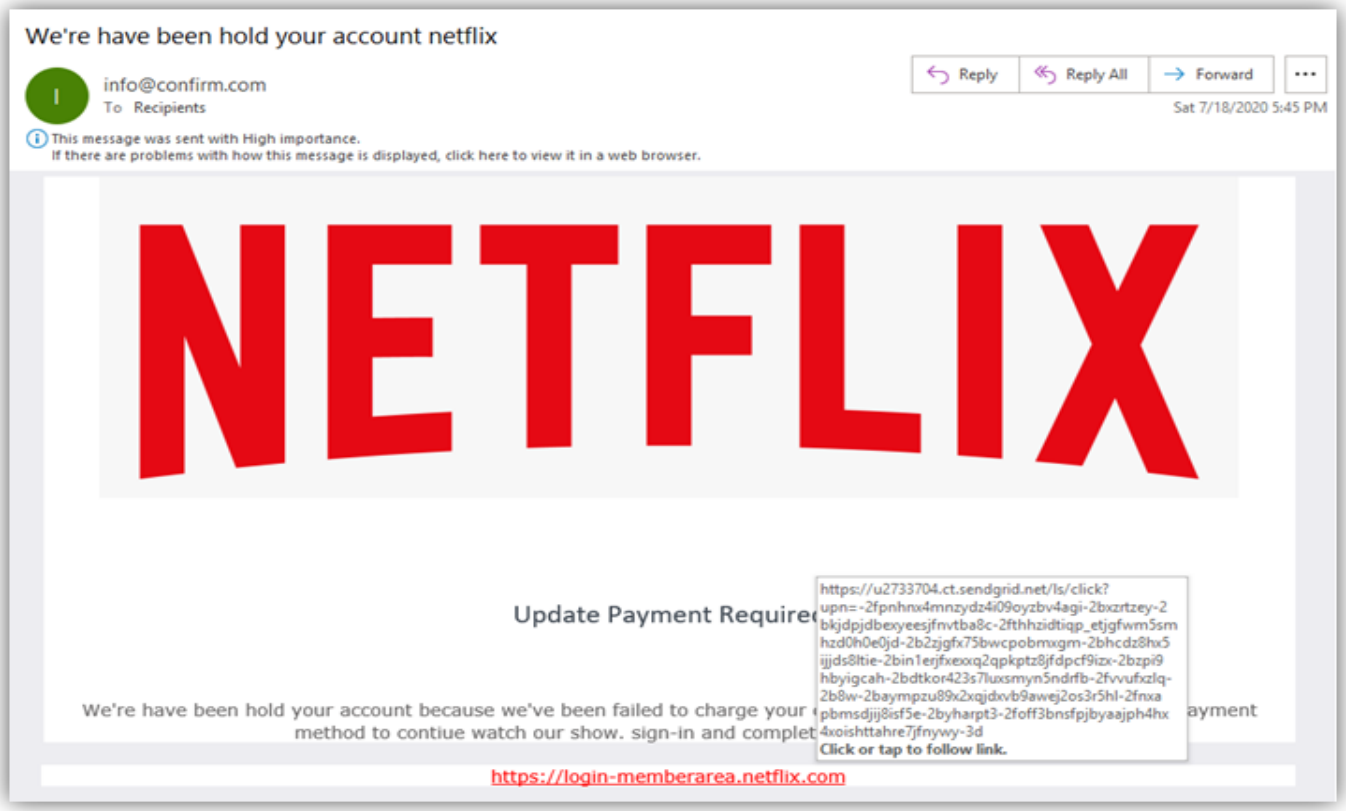
- Official Netflix login domain is:
  - https://www.netflix.com
- The domain shown:
  - login-memberarea.netflix.com
- While it appears similar, it is a **subdomain trick** used in phishing.

Additionally, when hovering over the button, the preview shows:

u273704.ct.sendgrid.net/...

This indicates redirection via third-party tracking or malicious routing.

Red Flag Identified: Domain spoofing and redirect-based phishing.



3 Check Email Header (Technical Inspection)

Although the full header is not shown in the screenshot, visible indicators include:

- This message was sent with High importance.
- Suspicious sender infrastructure.
- Use of external email distribution service (SendGrid redirect).

Professional Observation:

Phishing emails often:

- Use compromised SMTP servers
- Spoof sender display name
- Use bulk mailing services

Red Flag Identified: Email routing inconsistency with legitimate Netflix infrastructure.

4 Inspect Embedded Links

The call-to-action button says:

"Update Payment Requirement" Hover

preview shows:

u273704.ct.sendgrid.net/...

Analysis:

- Link does not directly go to netflix.com
- It redirects via tracking domain
- Final visible link: login-memberarea.netflix.com

This technique is called:

Credential Harvesting via Redirected Phishing Link

Red Flag Identified: Suspicious redirection chain.

5 Look for Urgency Language

The email states:

We've have been hold your account because we've been failed to charge your payment

method.

Analysis:

- Claims account suspension
- Mentions payment failure
- Pushes user to act

immediately Phishing psychology

tactic used:

f**ear + Urgency Manipulation**

This pressures victims into clicking without verifying.

Red Flag Identified: Artificial urgency to bypass rational thinking.

6 Check Grammar and Language Errors

Examples from the email:

- We've have been hold your account
- Incorrect verb usage
- Awkward sentence structure

Professional Assessment:

Legitimate corporate emails undergo:

- Legal review
- Brand consistency checks
- Grammar verification

Such errors strongly indicate fraudulent origin.

Red Flag Identified: Poor grammar inconsistent with professional brand communication.

Final Assessment

Based on:

- Sender domain mismatch
- Suspicious redirection links
- Urgency-based manipulation
- Grammar mistakes

- Infrastructure inconsistencies

This email is clearly identified as a **Phishing Attack Attempt** targeting user credentials and payment information.

1 Check Sender Email Address

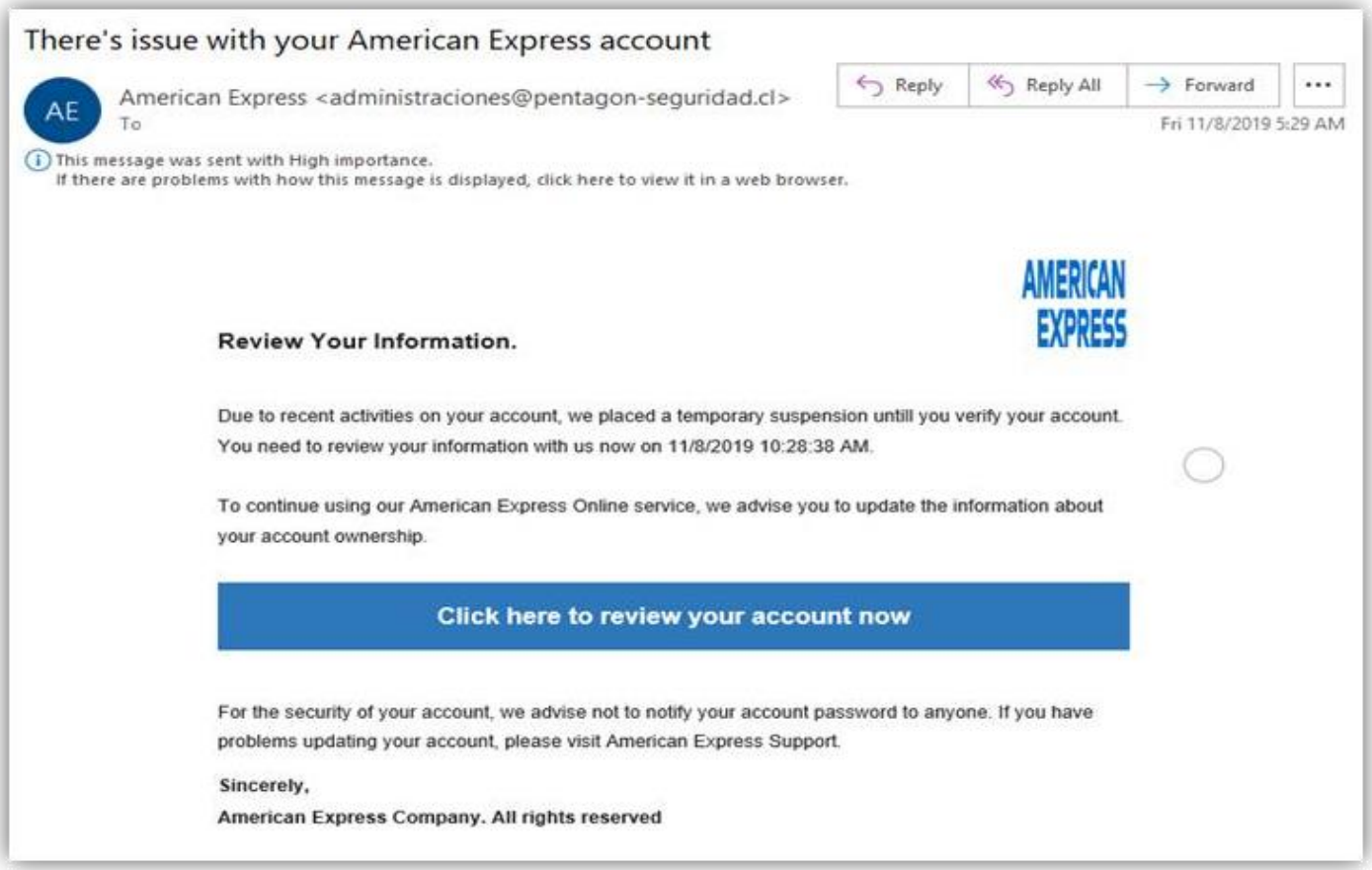
Sender shown:

administraciones@pentagon-seguridad.cl

Analysis:

- Official American Express domain:
  - @americanexpress.com
- Sender domain:
  - pentagon-seguridad.cl
- .clindicates Chile-based domain unrelated to American Express.

Critical Red Flag:
Domain mismatch → Clear brand impersonation attempt.


There's issue with your American Express account

AE American Express <administraciones@pentagon-seguridad.cl>
To

Reply | Reply All | Forward | ...
Fri 11/8/2019 5:29 AM

This message was sent with High importance.
If there are problems with how this message is displayed, click here to view it in a web browser.

AMERICAN EXPRESS

Review Your Information.

Due to recent activities on your account, we placed a temporary suspension untill you verify your account. You need to review your information with us now on 11/8/2019 10:28:38 AM.

To continue using our American Express Online service, we advise you to update the information about your account ownership.

Click here to review your account now

For the security of your account, we advise not to notify your account password to anyone. If you have problems updating your account, please visit American Express Support.

Sincerely,
American Express Company. All rights reserved

2 Verify Domain Legitimacy

The email claims to be from **American Express**, but:

- Domain is not owned by American Express.
- No visible corporate domain alignment.
- Likely spoofed display name:

"American Express" (Display Name Spoofing)

This is called:

Display Name Spoofing Attack

Where attacker shows trusted brand name but uses fake backend email.

Red Flag: Fake sender infrastructure.

3 Check Email Header (Technical Perspective)

Visible indicators:

- Marked as **High Importance**
- Generic structure
- No official footer verification links
- No official privacy/legal disclaimer format In real

American Express emails:

- SPF, DKIM, DMARC authentication passes
- Consistent legal footer
- Official domain hyperlinks

Likely header would show:

- Third-party SMTP relay
- Domain authentication failure

4 Inspect Embedded Links

Call-to-action button:

"Click here to review your account now"

Problem:

- Button link not shown clearly
- Likely redirects to phishing login page
- Classic **Credential Harvesting Page**

Financial phishing emails typically:

- Clone official login page
- Capture username/password
- Capture OTP or card details

High Risk Indicator: Financial brand + login verification request.

5 Look for Urgency Language

Email states:

We placed a temporary suspension until you verify your account. Review your information now.

Psychological manipulation technique:

- Account suspension threat
- Time-based urgency
- Fear trigger

This is known as:

Account Suspension Phishing Tactic

Designed to override rational verification.

6 Check Grammar and Content Quality

Issues noticed:

- “There's issue with your American Express account” (Missing “an”)
- Slightly unnatural phrasing
- Generic wording
  No personalization (no full name)

- Legitimate financial institutions:

- Address customer by full registered name
- Include masked account numbers
- Use consistent brand formatting

Poor grammar + generic message = phishing probability high.

Final Security Assessment

Based on:

- Sender domain mismatch
- Foreign unrelated domain (.cl)
- Urgency-based manipulation
- Suspicious call-to-action button
- Grammar issues
- Financial credential request

This email is a **High-Risk Financial Phishi**

REMINDER: Export Documents//Draft B/L # DOVUN4873

ML Maersk Line <Lychheng.Ngor@Ins.Maersk.co.cn>
To

If there are problems with how this message is displayed, click here to view it in a web browser.

DOVUN4873.HTML
822 bytes

Action Items

Dear ,

Attached you will find a copy of the stamped bill of lading and the notification of arrival for the cargo that is expected on the aforementioned date ETA: March 30, 2020.

Best Regards,

Lychheng Ngor

Customer Service Associate
Customer Service
Maersk SCM

MAERSK LINE

Damco (Cambodia) Ltd.
VTrust Tower - 7th Floor,
#Plot A, Street 169, Phum 12,
Sangkat Veal Vong, Khan 7 Makara,
Phnom Penh,

7 Sample Email Analysis

Email 1 – Phishing (Netflix Account Suspension)

- **Sender:**
info@confirm.com

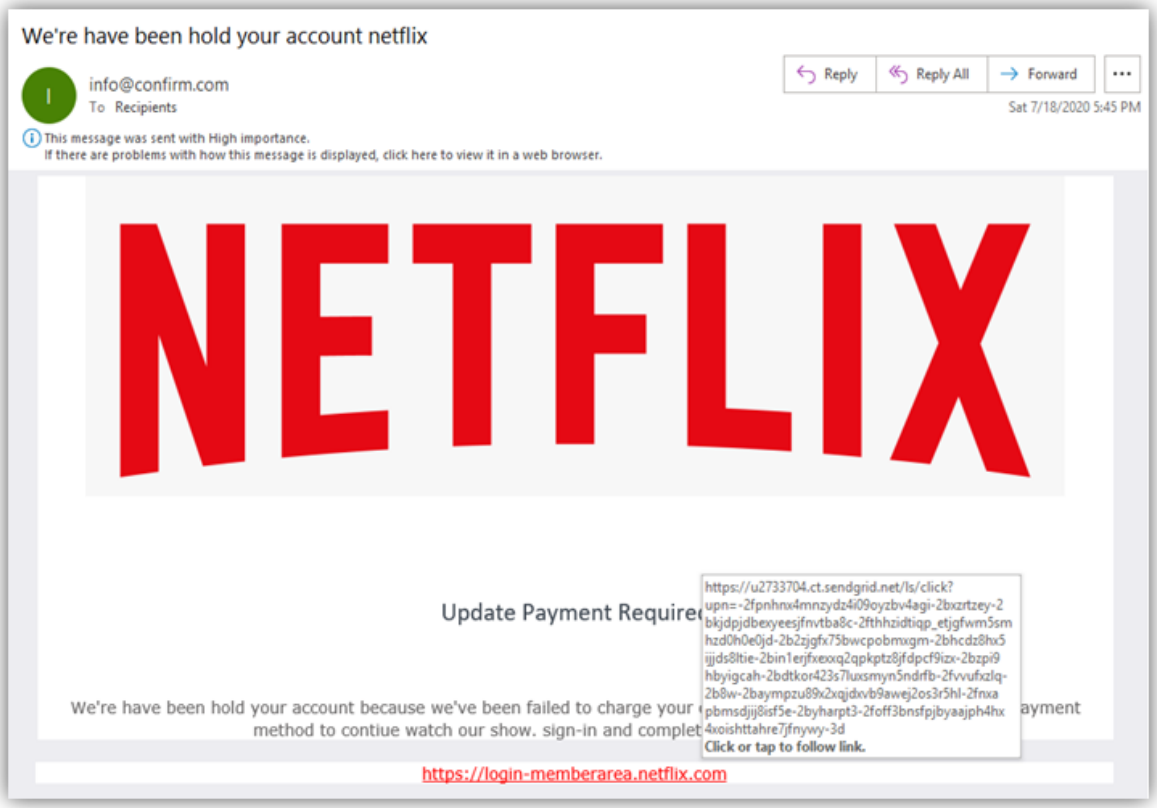
- **Subject:**
We're have been hold your account netflix

- **Indicators:**

  - Suspicious sender domain (not netflix.com)
  - Grammar mistake in subject line
  - Fake login link (login-memberarea.netflix.com)
  - Urgency related to payment failure
  - Branded logo misuse
  - Redirect link via third-party tracking (SendGrid)

Risk Level:
High Risk

- **Reason:**
The email attempts credential harvesting by impersonating Netflix and redirecting users to a fake login portal. The sender domain does not belong to Netflix, and grammatical errors further indicate phishing intent.



Email 2 – Phishing (American Express)

- **Sender:**

administraciones@pentagon-seguridad.cl

- **Subject:**

There's issue with your American Express account.

- **Indicators:**

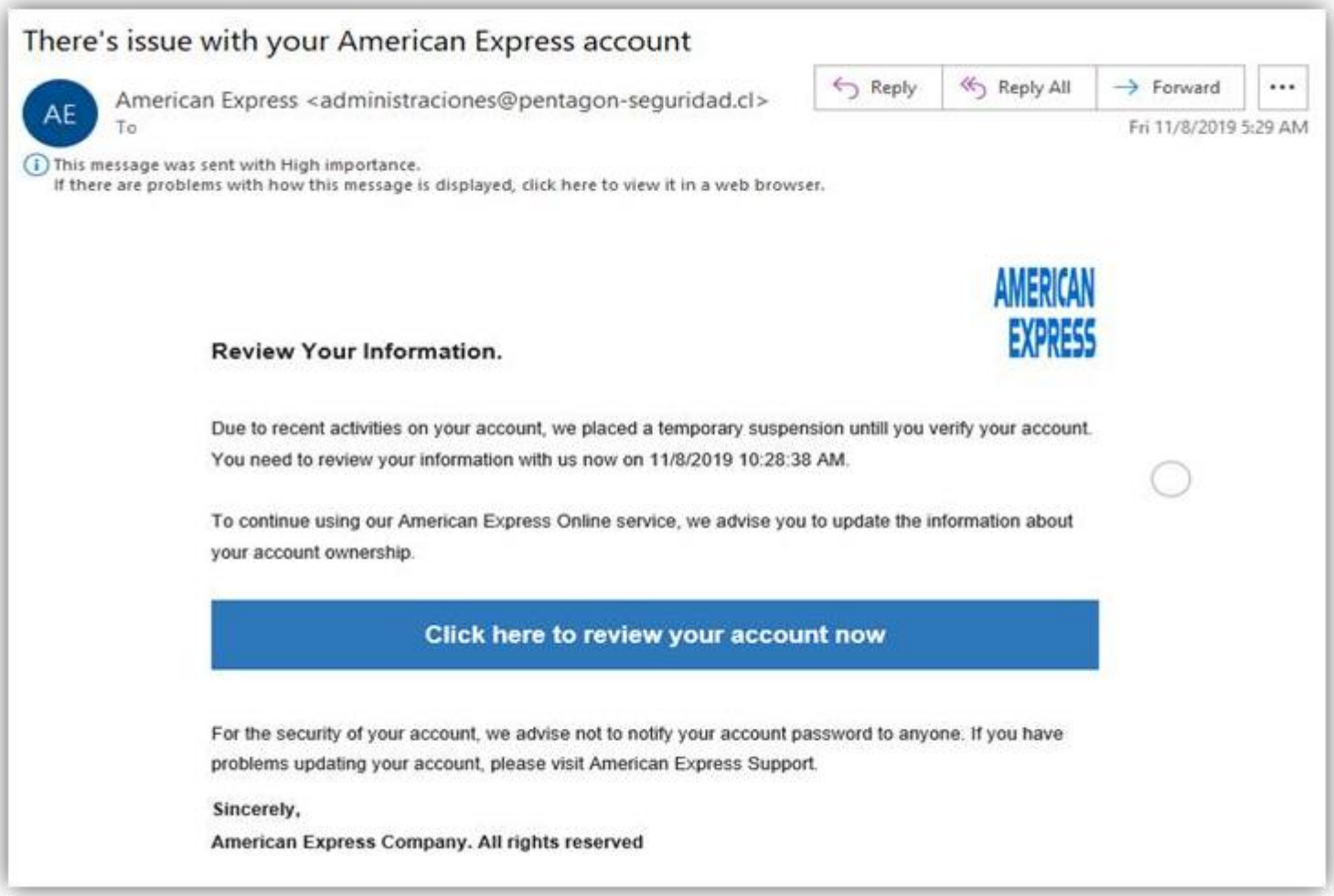
  - Sender domain unrelated to American Express
  - Domain impersonation
  - Urgency regarding account suspension
  - Generic call-to-action button
  - No personalization
  - Financial data verification request

- **Risk Level:**

High Risk

- **Reason:**

The email impersonates American Express but originates from a suspicious domain. It attempts to trick users into verifying account information through a malicious link.



There's issue with your American Express account

AE American Express <administraciones@pentagon-seguridad.cl>
To

Reply | Reply All | Forward | ...

Fri 11/8/2019 5:29 AM

This message was sent with High importance.
If there are problems with how this message is displayed, click here to view it in a web browser.

AMERICAN EXPRESS

Review Your Information.

Due to recent activities on your account, we placed a temporary suspension untill you verify your account. You need to review your information with us now on 11/8/2019 10:28:38 AM.

To continue using our American Express Online service, we advise you to update the information about your account ownership.

Click here to review your account now

For the security of your account, we advise not to notify your account password to anyone. If you have problems updating your account, please visit American Express Support.

Sincerely,
American Express Company. All rights reserved

Email 3 – Business Email Compromise (Maersk)

- **Sender:**
Lychheng.Ngor@Ins.Maersk.co.cn

- **Subject:**
REMINDER: Export Documents/Draft B/L # DOVUN4873

- **Indicators:**
  - Look-alike domain (maersk.co.cn instead of maersk.com)
  - HTML attachment (DOVUN4873.HTML)
  - Blank greeting (“Dear ,”)
  - Business-themed urgency
  - Shipping/invoice lure

- **Risk Level:**
High Risk

- **Reason:**
The email appears to be a Business Email Compromise attempt using a spoofed domain and malicious HTML attachment to deliver phishing content or malware.

REMINDER: Export Documents//Draft B/L # DOVUN4873

ML Maersk Line <Lychheng.Ngor@Ins.Maersk.co.cn>
To

If there are problems with how this message is displayed, click here to view it in a web browser.

DOVUN4873.HTML
822 bytes

Action Items

Dear ,

Attached you will find a copy of the stamped bill of lading and the notification of arrival for the cargo that is expected on the aforementioned date ETA: March 30, 2020.

Best Regards,

Lychheng Ngor

Customer Service Associate
Customer Service
Maersk SCM

MAERSK LINE

Damco (Cambodia) Ltd.
VTrust Tower - 7th Floor,
#Plot A, Street 169, Phum 12,
Sangkat Veal Vong, Khan 7 Makara,
Phnom Penh,

Email 4 – Advance Fee Scam (Google Forms Fraud)

- **Sender:**
alhashimyreem7@gmail.com

- **Subject:**
Hello

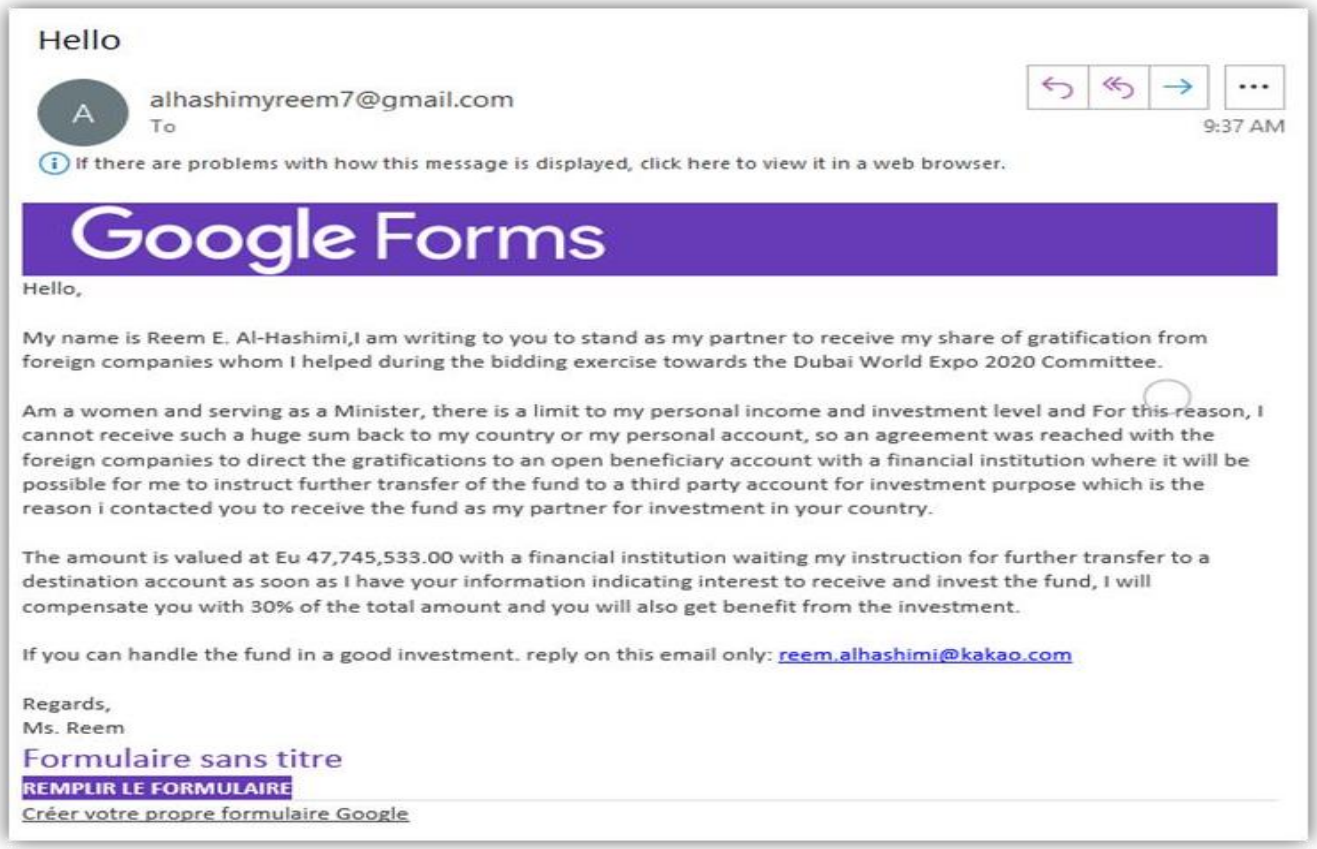
- **Indicators:**

  - Free email service (Gmail)
  - Large financial reward promise (€47 million)
  - Investment partnership request
  - 30% commission offer
  - Unrealistic financial narrative
  - Emotional manipulation

- **Risk Level:**
High Risk

- **Reason:**

This is a classic advance-fee fraud scam offering a share of large funds in exchange for cooperation, typically designed to extract money or personal information.



Hello

A alhashimyreem7@gmail.com
To

9:37 AM

If there are problems with how this message is displayed, click here to view it in a web browser.

Google Forms

Hello,

My name is Reem E. Al-Hashimi,I am writing to you to stand as my partner to receive my share of gratification from foreign companies whom I helped during the bidding exercise towards the Dubai World Expo 2020 Committee.

Am a women and serving as a Minister, there is a limit to my personal income and investment level and For this reason, I cannot receive such a huge sum back to my country or my personal account, so an agreement was reached with the foreign companies to direct the gratifications to an open beneficiary account with a financial institution where it will be possible for me to instruct further transfer of the fund to a third party account for investment purpose which is the reason i contacted you to receive the fund as my partner for investment in your country.

The amount is valued at Eu 47,745,533.00 with a financial institution waiting my instruction for further transfer to a destination account as soon as I have your information indicating interest to receive and invest the fund, I will compensate you with 30% of the total amount and you will also get benefit from the investment.

If you can handle the fund in a good investment. reply on this email only: reem.alhashimi@kakao.com

Regards,
Ms. Reem

Formulaire sans titre

REMPLIR LE FORMULAIRE

Créer votre propre formulaire Google

Email 5 – Malware Delivery (Voicemail Scam)

- **Sender:**

Unknown/External Domain

- **Subject:**

New VoiceMail from +1 (502)-564-6546"

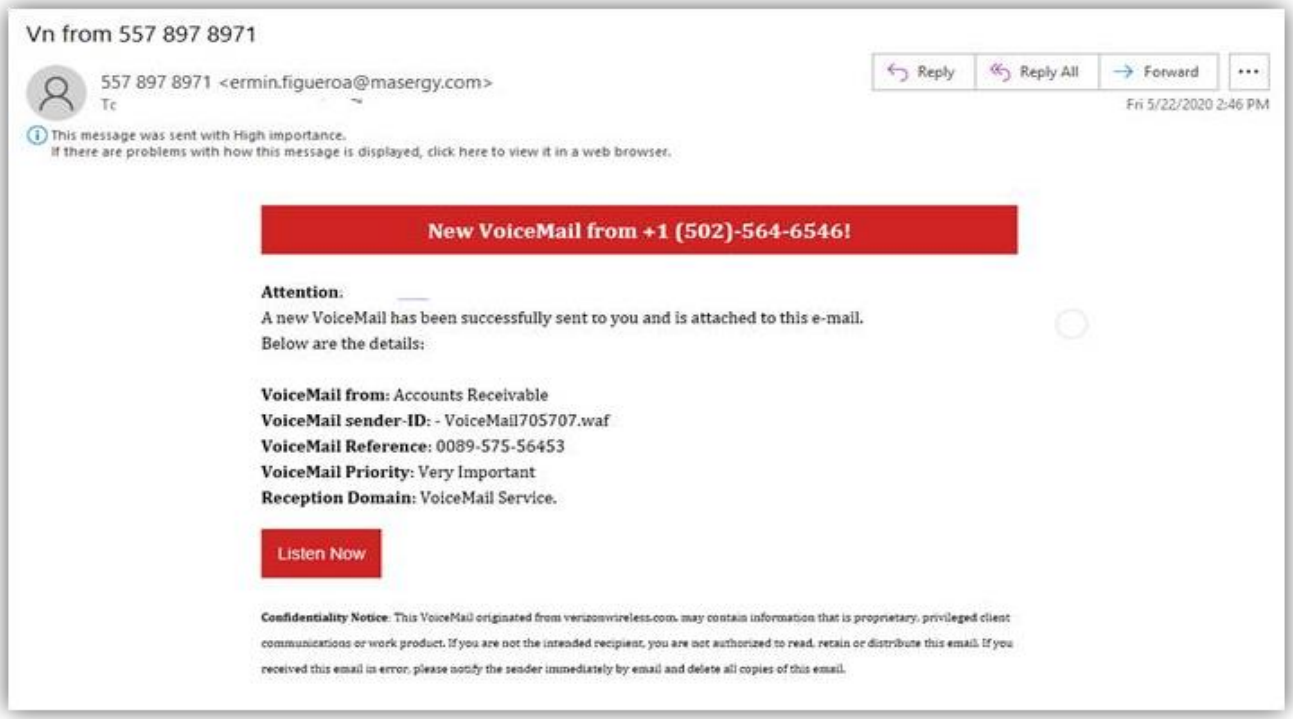
- **Indicators:**

  - Unexpected voicemail notification
  - Suspicious attachment (.wav file)
  - Generic message
  - No prior context
  - High importance flag

- **Risk Level:**

High Risk

- **Reason:**
The email likely contains a malicious attachment disguised as voicemail audio, commonly used to distribute malware or redirect users to phishing portals.


Vn from 557 897 8971

557 897 8971 <ermin.figueroa@masergy.com>
To

Reply | Reply All | Forward | ...

Fri 5/22/2020 2:46 PM

This message was sent with High importance.
If there are problems with how this message is displayed, click here to view it in a web browser.

New VoiceMail from +1 (502)-564-6546!

Attention:
A new VoiceMail has been successfully sent to you and is attached to this e-mail.
Below are the details:

VoiceMail from: Accounts Receivable
VoiceMail sender-ID: - VoiceMail705707.waf
VoiceMail Reference: 0089-575-56453
VoiceMail Priority: Very Important
Reception Domain: VoiceMail Service.

Listen Now

Confidentiality Notice: This VoiceMail originated from verizonwireless.com, may contain information that is proprietary, privileged client communications or work product. If you are not the intended recipient, you are not authorized to read, retain or distribute this email. If you received this email in error, please notify the sender immediately by email and delete all copies of this email.

Phishing Indicators Identified

Spoofed Domain – The sender's email domain does not match the official domain of the organization (e.g., *confirm.com* instead of *netflix.com*, *pentagon- seguridad.cl* instead of *americanexpress.com*).

Urgent Tone – The email uses urgency tactics such as account suspension, payment failure warnings, or document deadlines to pressure the recipient into taking immediate action.

Suspicious Link – The hyperlink appears to represent a legitimate brand; however, the actual URL redirects to a different or unrelated domain.

Malicious Attachment Risk – Attachments such as HTML or WAV files may contain malicious scripts or be used for credential harvesting attacks.

Grammar and Language Errors – The subject line and email body contain grammatical mistakes (e.g., "We're have been hold…", "There's issue…"), which are uncommon in legitimate corporate communications.

Low Risk

Definition:
An email or message that shows minor suspicious indicators but does not contain direct malicious links or harmful attachments.

Characteristics:

- Slight domain mismatch
- Generic greeting (e.g., "Dear User")
- Minor formatting issues
- No clickable link or attachment

Impact:
Limited immediate threat. However, it should still be reported and monitored.

Example:
Marketing-style emails from unknown domains with no active phishing payload.

Medium Risk

Definition:
An email containing suspicious links or attachments but lacking strong evidence of active malware or credential harvesting.

Characteristics:

- Suspicious hyperlink redirect
- Unexpected attachment (HTML, PDF, DOC)
- Urgency language
- Brand impersonation

Impact:
Moderate risk of credential theft or malware download if the user interacts.

Example:
Fake invoice emails or document-sharing notifications.

High Risk

Definition:
An email clearly designed to steal credentials, distribute malware, or cause financial fraud.

Characteristics:

- Spoofed sender domain
- Credential harvesting login page
- Malicious attachments (HTML, executable, macro-enabled files)
- Strong urgency or threat language
- Multiple phishing indicators present

Impact:
Severe risk including:

- Account compromise
- Financial loss
- Data breach
- Malware infection

Example:
Fake Netflix / American Express login page asking for password update.

10 Impact on Organization

Phishing attacks can have serious consequences for organizations, affecting financial stability, data security, reputation, and regulatory compliance. The potential impacts include:

Financial Loss

Successful phishing attacks can result in direct financial losses through fraudulent transactions, unauthorized fund transfers, or ransomware payments. In addition, organizations may incur indirect costs such as incident response expenses, system recovery costs, and business downtime.

Data Breach

Phishing often leads to unauthorized access to sensitive corporate data, including customer information, employee records, intellectual property, and financial documents. A single compromised credential can allow attackers to move laterally within systems and escalate the breach.

Reputational Damage

When customers or partners learn that an organization has suffered a phishing- related breach, trust can be significantly reduced. Loss of customer confidence may impact long-term business relationships, brand credibility, and market value.

Legal and Regulatory Penalties

Organizations handling personal or financial data are subject to data protection regulations. A phishing-induced data breach may lead to regulatory investigations, compliance penalties, legal actions, and contractual liabilities

1 Prevention Guidelines

To reduce the risk of phishing attacks, organizations should implement the following preventive security measures:

Enable Multi-Factor Authentication (MFA)

Implement Advanced Email Filtering

Conduct Regular Security Awareness Training

Configure SPF, DKIM, and DMARC

Avoid Clicking Unknown or Suspicious Links

12 Awareness Guidelines for Employees

Employees play a critical role in preventing phishing attacks, as most phishing attempts target human behavior rather than technical systems. The following awareness guidelines should be followed:

Carefully Verify the Sender

Hover Over Links Before Clicking

Be Cautious with Attachments

Do Not React to Urgency Immediately

Never Share Sensitive Information

Report Suspicious Emails Immediately

Conclusion

The analysis of the selected email samples demonstrates how phishing attacks exploit domain spoofing, malicious links, suspicious attachments, urgency tactics, and grammatical inconsistencies to deceive users. Through structured evaluation and risk classification, it is evident that phishing remains a significant threat to both individuals and organizations. Implementing technical controls alongside continuous user awareness is essential to reduce the likelihood of successful phishing attacks and protect sensitive information.

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